NaSCon '26 App Dev Competition

Official Rules, Environment Readiness & Compliance Booklet

Dates: Oct 17–18, 2026 | **Venues:** Day 1 – A01 to A03, A103 and A104 | Day 2 – C-401 to C-405, FAST-NUCES Islamabad

Prize Pool: Winner: PKR 35,000 | Runner-Up: PKR 15,000 | **Team Size:** 1–3 Members

1. Competition Schedule & Two-Phase Structure (Strict 8 Hours)

The competition is divided into two locked phases across two days. Working overnight is impossible because Phase I is graded and permanently closed at 15:00 on Day 1, while Phase II is kept confidential until 10:15 on Day 2.

- **Day 1 (Saturday, Oct 17):** Check-in and empty repository check: 10:00–10:44 AM. **Sprint 1 (Part I):** 10:45 AM–03:00 PM (4 hours 15 minutes). **Day 1 Cloud Autograder Audit:** 03:05–03:30 PM (40 points — permanently locked).
- **Day 2 (Sunday, Oct 18):** Morning commit verification: 10:00–10:14 AM. **Sprint 2 (Part II):** 10:15 AM–02:00 PM (3 hours 45 minutes). **Day 2 Cloud Autograder Audit:** 02:05–02:25 PM (45 points). **Top-8 Finalist Viva:** 02:30–03:50 PM (15 points).

2. Mandatory Pre-Pulled Docker Bundle (Run at Home)

Large downloads and Docker Hub image pulls are throttled on campus Wi-Fi. To keep both Phase I and Phase II problem domains strictly confidential, all teams must download the required Docker images before arriving on campus.

The bundle is called the **NaSCon Universal Full-Stack & QA Image Bundle**. Some images are used in Phase I, some in Phase II, and some are decoys.

Run the following commands at home before Saturday at 09:00 AM.

```
# Download these images before arriving on campus.
docker pull postgres:16-alpine
docker pull redis:7-alpine
docker pull minio/minio:RELEASE.2024-09-22T00-33-43Z
docker pull rabbitmq:3.13-management-alpine
docker pull nats:2.10-alpine
docker pull bluenviron/mediamtx:1.9.1
docker pull livekit/livekit-server:v1.7.2
docker pull binwiederhier/ntfy:v2.11.0
docker pull prom/prometheus:v2.54.1
docker pull grafana/grafana:11.2.0
docker pull grafana/k6:0.54.0
docker pull selenium/hub:4.24.0
docker pull selenium/node-chrome:4.24.0
```

```
docker pull selenium/node-firefox:4.24.0

# Intel/AMD Windows and Linux laptops:
docker pull selenium/node-edge:4.24.0

# Apple Silicon Macs should use:
# seleniarm/node-chromium:4.24.0

# Also download ONE backend runtime image.
# Examples:
docker pull python:3.12-slim
# OR
docker pull node:20-alpine
```

2.1 Mobile Platform, QA Tools & Private Hotspot Requirement

1. **Mobile Platform (One Is Sufficient):**

   Supporting either Android or iOS satisfies the mobile platform requirement. Teams may use Flutter, React Native, native Android development with Kotlin or Java, or native iOS development with Swift.

   Download and cache the required Gradle, CocoaPods, and Pub dependencies before the competition so they are available offline.

   Bringing a physical phone with a USB data cable is strongly recommended. Enable USB debugging where applicable. Using a physical phone instead of an emulator can save approximately 4 GB of RAM.

2. **Automated Testing Tools:**

   Install Appium 2 with either the UiAutomator2 driver for Android or the XCUITest driver for iOS.

   Also install Selenium WebDriver bindings and the required testing framework, such as pytest, JUnit, or TestNG. Configure Allure or HTML report generation and install either k6 or Locust for performance testing.

3. **Phone Hotspot or Travel Router for the Team Network:**

   Campus Wi-Fi isolates connected devices and prevents them from communicating directly with one another.

   Bring a smartphone that can create a local Wi-Fi hotspot or a travel router. This allows team laptops and the test phone to communicate over a local network using their local IP addresses.

4. **Campus Wi-Fi Allowlist:**

   Campus Wi-Fi is used for pushing code to GitHub, running GitHub Actions, accessing AI assistants such as ChatGPT, Claude, Cursor, and Copilot, and downloading lightweight packages.

   The required package sources include `pypi.org`, `registry.npmjs.org`, `pub.dev`, and `repo.maven.apache.org`.

3. Low-RAM (8 GB / 16 GB) Laptop Survival Guide

- **Cloud Grading Reduces the Load on Your Laptop:**

  At the submission deadlines (03:00 PM on Day 1 and 02:00 PM on Day 2), the official GitHub Actions Autograder runs your `docker-compose.yml` file on GitHub's cloud runner, which has 16 GB of RAM and uses `ubuntu-latest`.

  Your laptop does not need to run every container simultaneously during the official grading process.

- **Split Local Testing Across Team Members:**

  During the coding sprint, teams using laptops with 8 GB or 16 GB of RAM should distribute local testing workloads across teammates' machines connected through the team's phone hotspot.

  A suggested arrangement is:

  - **Laptop 1:** Backend services and containers.
  - **Laptop 2:** Mobile application and physical phone connected through USB.
  - **Laptop 3:** Selenium browser containers and automated tests.

  To reduce memory usage, configure browser containers with `shm_size: "1gb"` and `SE_START_VNC: "false"`.

  Here is what these settings mean:

  - `shm_size: "1gb"` allocates 1 GB of shared memory to the container. Shared memory is used by processes running inside the container, including browser processes.
  - `SE_START_VNC: "false"` disables the VNC service that provides a remotely viewable browser desktop in Selenium browser containers.

  These settings are intended to reduce unnecessary resource usage during local testing.

4. Zero-Overnight Integrity & Cloud Autograding Rules

1. **Personal GitHub Account and Empty Repository at 10:40 AM:**

   Create a private GitHub repository under the team lead's personal GitHub account, which provides 2,000 free GitHub Actions minutes, and invite `nascon-eval-26` to the repository.

   At 10:40 AM on Day 1, the repository must be completely empty, with zero commits.

   All project scaffolding, Dockerfiles, and source code must be committed after 10:45 AM.

2. **Hourly Commit Rule:**

   Teams must push their code to GitHub at least once every hour during the coding sprints.

   Each missed hourly push incurs a penalty of 2 points.

3. **GitHub Server Timestamps Determine Eligibility:**

   Only commits pushed to GitHub by 03:05:00 PM PKT on Day 1 and 02:05:00 PM PKT on Day 2 are eligible for grading.

Local files that have not been committed, Git stashes, and pushes made after the respective deadlines are ignored by the cloud autograder.

4. **Zero Overnight Commits:**

   Between 03:05 PM PKT on Day 1 and 10:15 AM PKT on Day 2, no commits may be pushed to GitHub.

   On Day 2, the repository history must continue directly from the locked Day 1 commit SHA (the unique identifier of the commit).

5. **AI Tools and Top-8 Live Mutation Viva:**

   AI tools, including Cursor, Claude, ChatGPT, and Copilot, are fully allowed.

   Managed backend-as-a-service platforms (BaaS), including Firebase and Supabase, are prohibited. Using pre-built repositories is also prohibited.

   After the Day 2 cloud audit, the eight highest-scoring teams advance to a 10-minute viva with the judge.

   The viva includes:

   - A 5-minute live code mutation task.
   - Two hidden complexity scenarios.
   - A defense of eight short Architecture Decision Records documented in `ADR.md`.
   - A defense of the traceability matrix documented in `TRACEABILITY.md`.

5. Tie-Breakers & Penalty Floor

All individual item scores and total scores are floored at zero. No score can fall below zero.

Any tie, including a tie for qualification among the Top 8 teams or for the first- and second-place prizes, is resolved using the following criteria in order:

1. Higher objective score on Day 2.
2. More automated test vectors passed across both days.
3. Earlier final GitHub push timestamp on Day 2.